

Multiple Choice

1. **Answer:** D
Options: A) 2; B) 3; C) 4; D) 5
Note: Correct option: D - 5
2. **Answer:** D
Options: A) Silicon; B) Diamond; C) Silicon carbide; D) Arsenic-doped silicon
Note: Correct option: D - Arsenic-doped silicon
3. **Answer:** D
Options: A) Na_2CO_3 ; B) Na_3PO_4 ; C) Na_2S ; D) NaCl
Note: Correct option: D - NaCl
4. **Answer:** D
Options: A) electron distribution; B) dipole moment; C) force constant; D) reduced mass
Note: Correct option: D - reduced mass
5. **Answer:** C
Options: A) 54.91 MHz; B) 239.2 MHz; C) 345.0 MHz; D) 2167 MHz
Note: Correct option: C - 345.0 MHz

True / False

6. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'Each hybrid orbital may hold four electrons.'
7. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'meso compound'.
8. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'Ca'.
9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $4.126 \times 10^{-3} \text{ at } 0.335T$; $2.142 \times 10^{-6} \text{ at } 10.5T$ '.

Long Form Response

11. **Answer:** Neither position nor momentum can be known exactly.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 9.4 mCi. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key